

GATE

CRASH COURSE

CS/IT

Data Structure & Programming

Lecture 05

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Topics *to be covered*

1

Tree construction using pre-order / Inorder
post order / Inorder

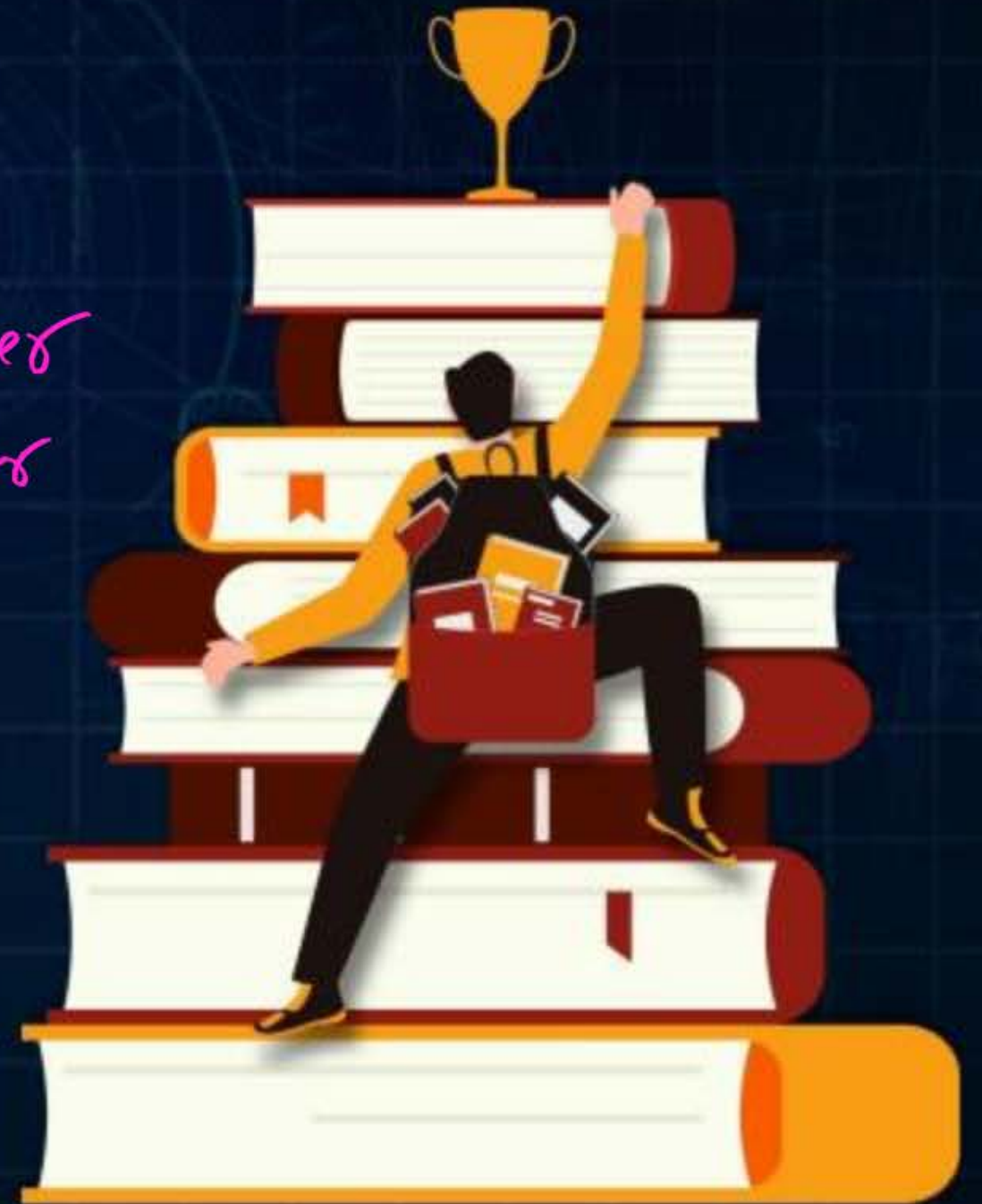
2

Binary Search tree construction

3

Search min max counting, traversal.

4





Building tree from pre- and in- order

Construction of binary tree from preorder and inorder traversal.

Root Left Right

• Preorder: $a b c d f g e$

Left Root Right

• Inorder: $c b f d g a e$

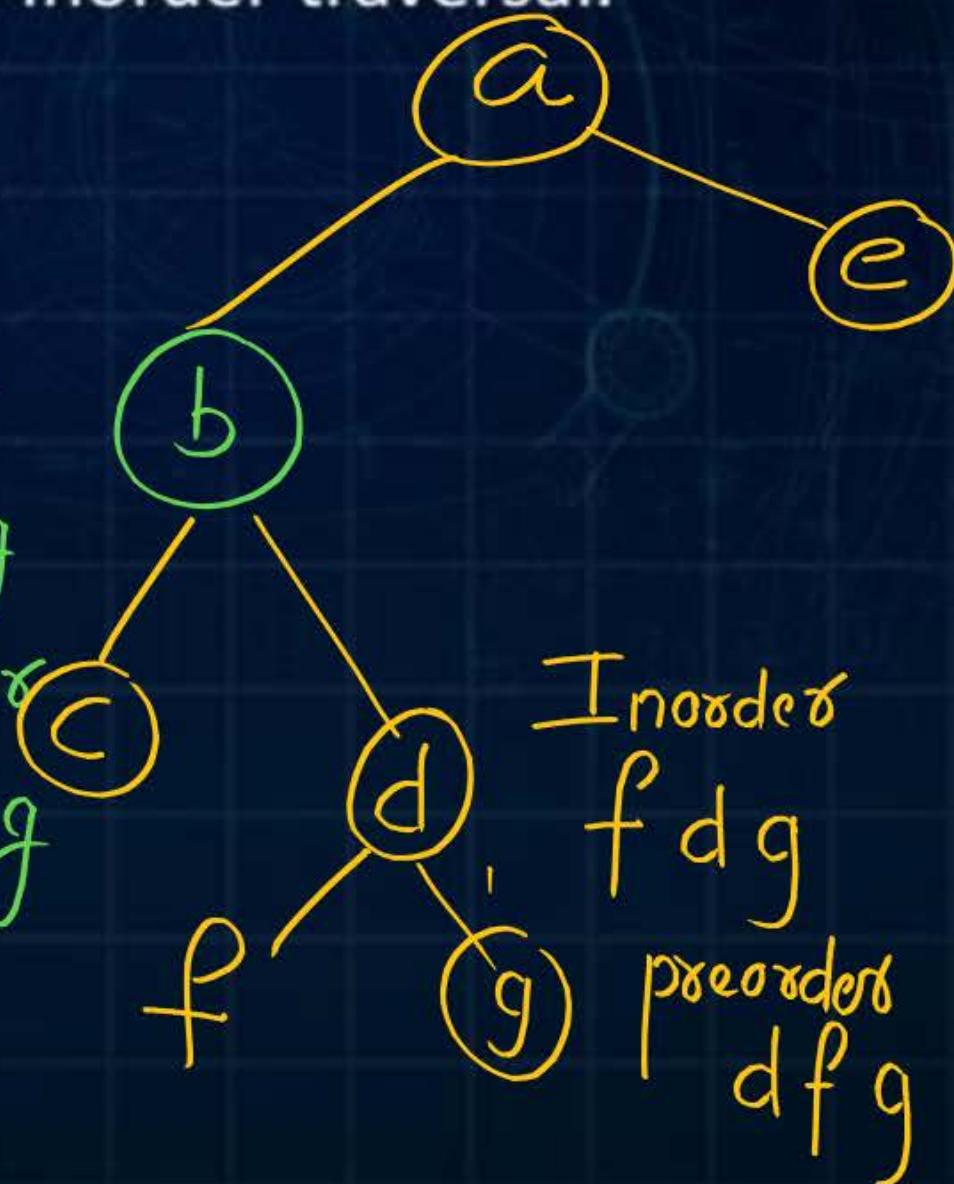
Identify the Root

Inorder Left and right
Subtree

preorder
 $b | c | d | f | g$

Inorder

$c b f d g$
Left





Building tree from post- and in- order

- Given the preorder and inorder traversals of a binary tree we can uniquely determine the tree.

Postorder:

Left Right Root

c f g d b e **a**

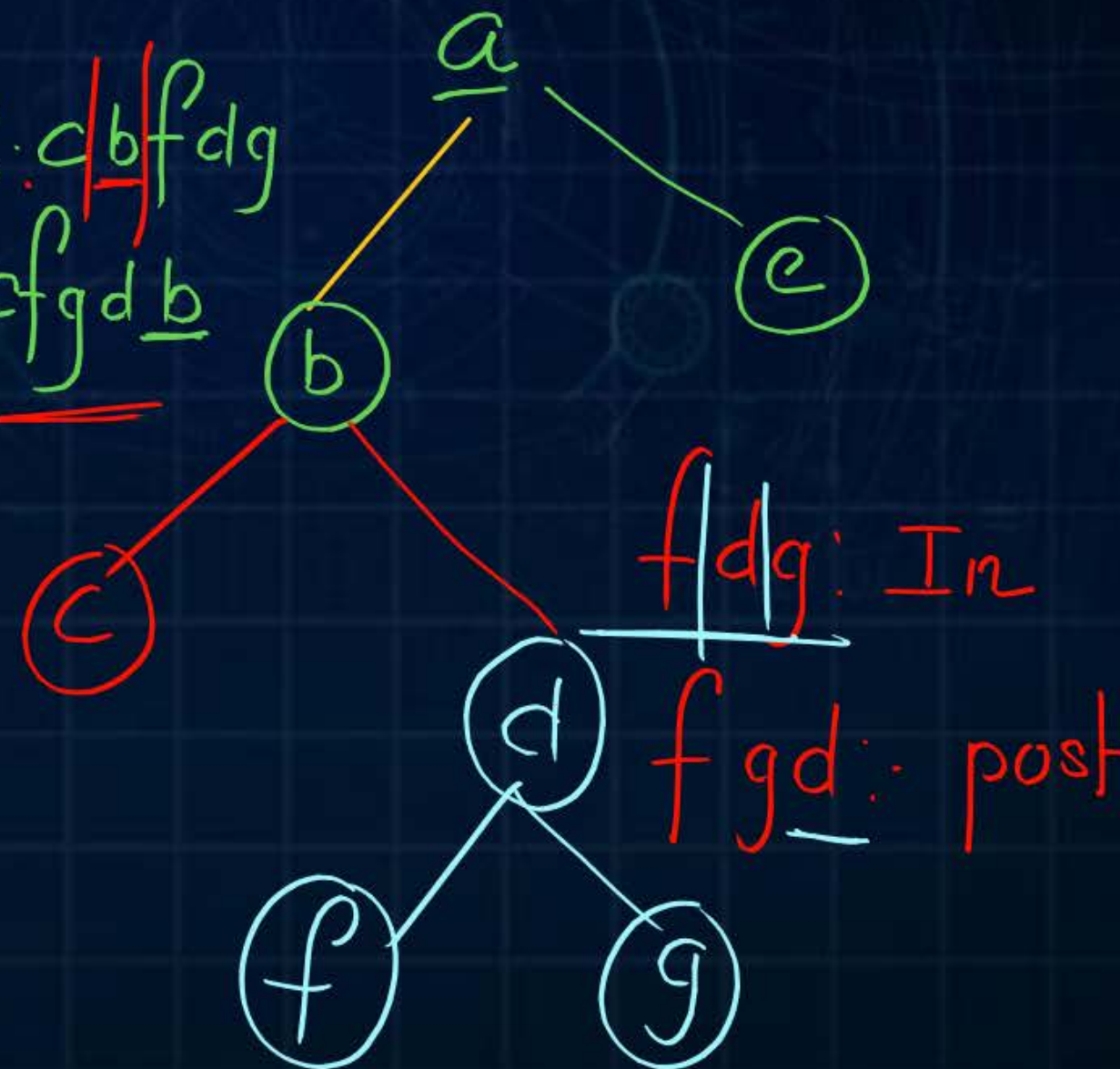
↑ Last element is
Root

Inorder

c b f d g **a** e

In: c b f d g

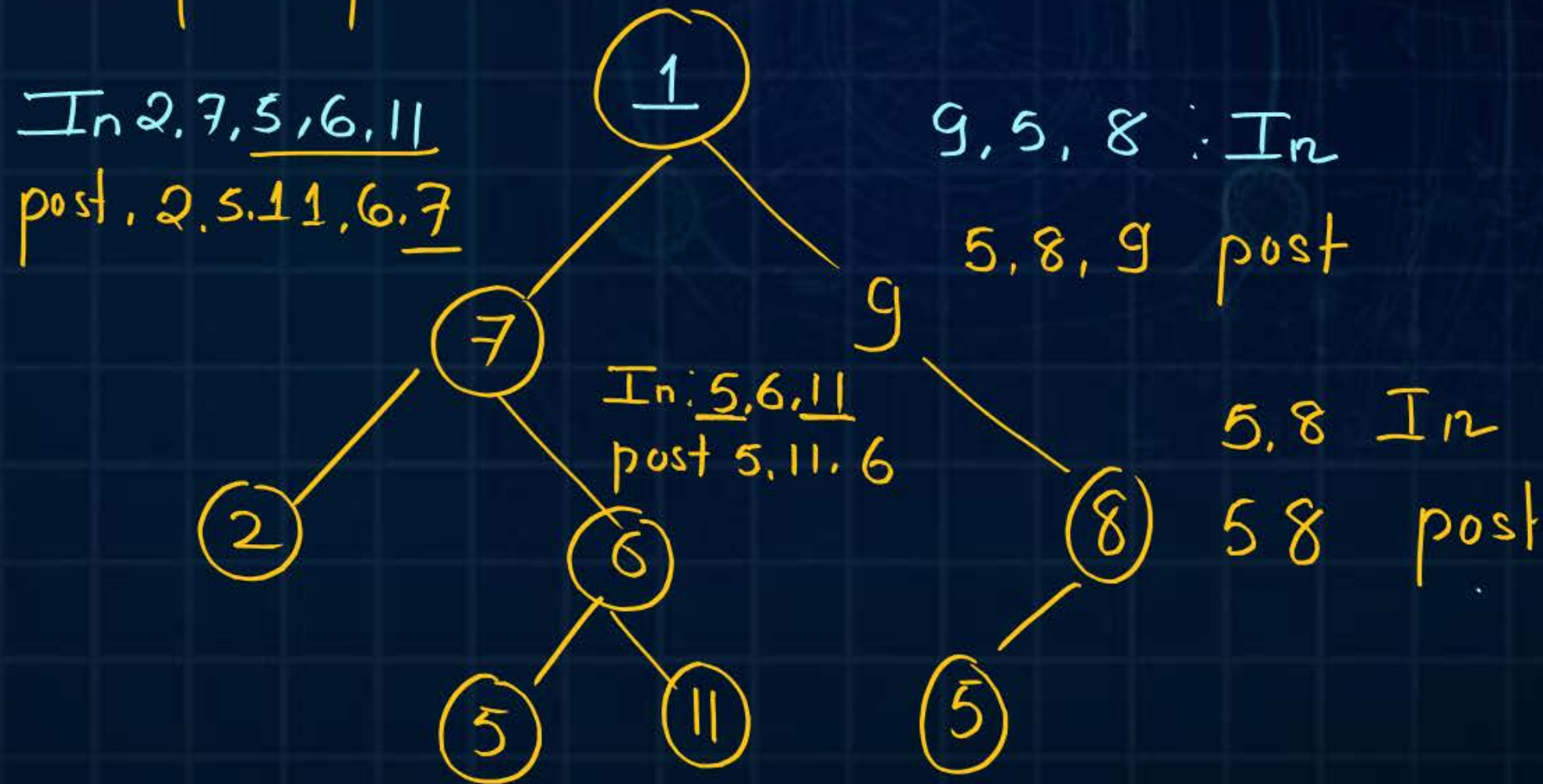
post: c f g d b



Question

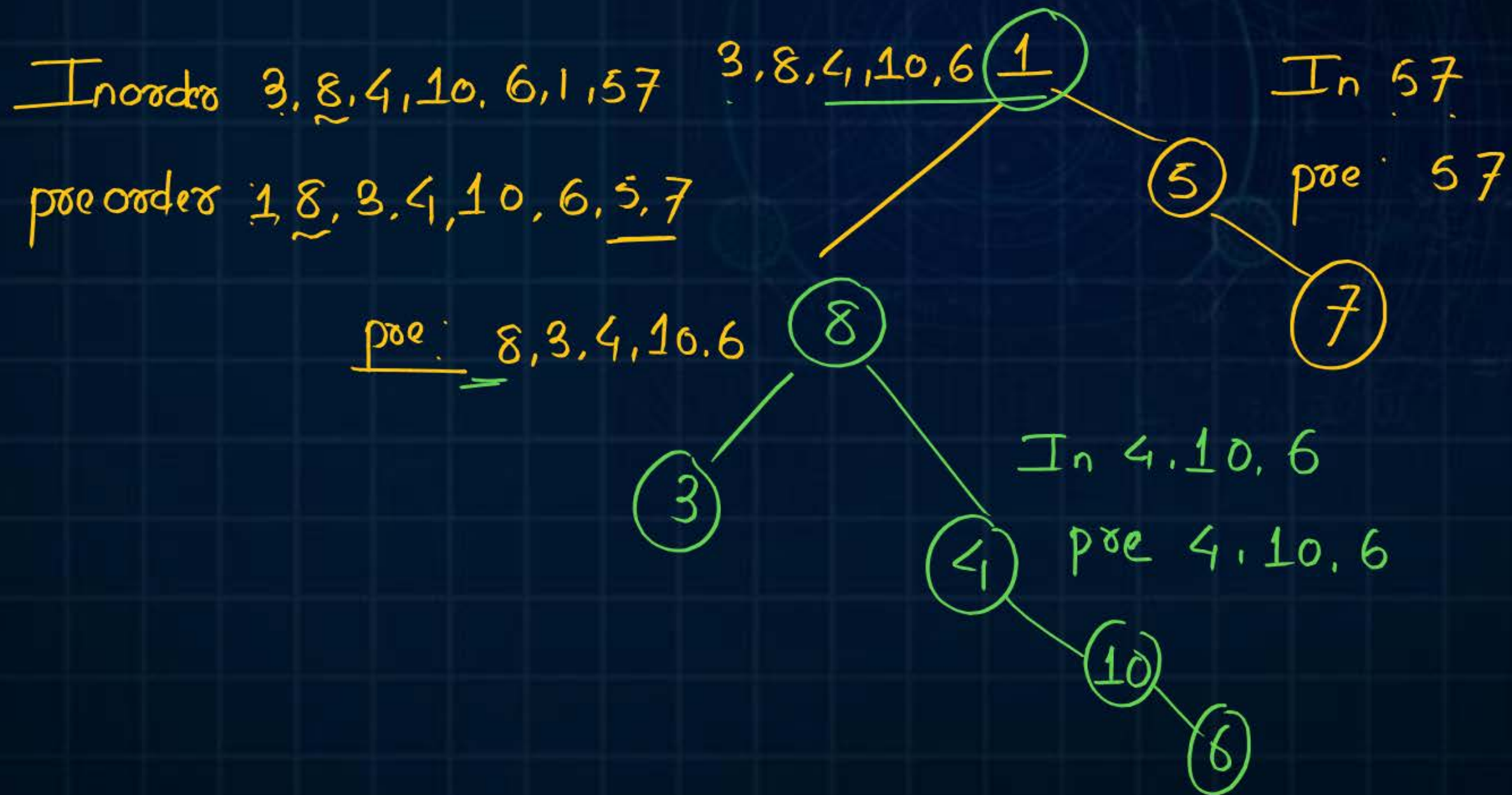
#Q if the inorder traversal is 2, 7, 5, 6, 11, 1, 9, 5, 8

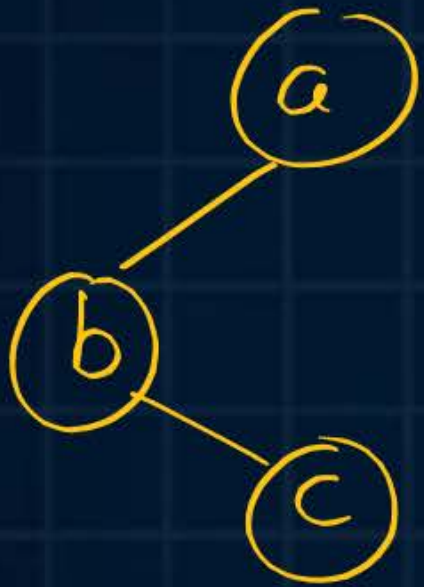
Post order : 2, 5, 11, 6, 7, 5, 8, 9, 1. If binary tree constructed then sum of leaf nodes is [23]



Question

#Q If the inorder traversal of a binary tree is 3, 8, 4, 10, 6, 1, 5, 7 and the preorder traversal is 1, 8, 3, 4, 10, 6, 5, 7 Then the sum of internal nodes is 28.





preorder: a b c
postorder: c b a



preorder: a b c
postorder: c b a

pre order
post order

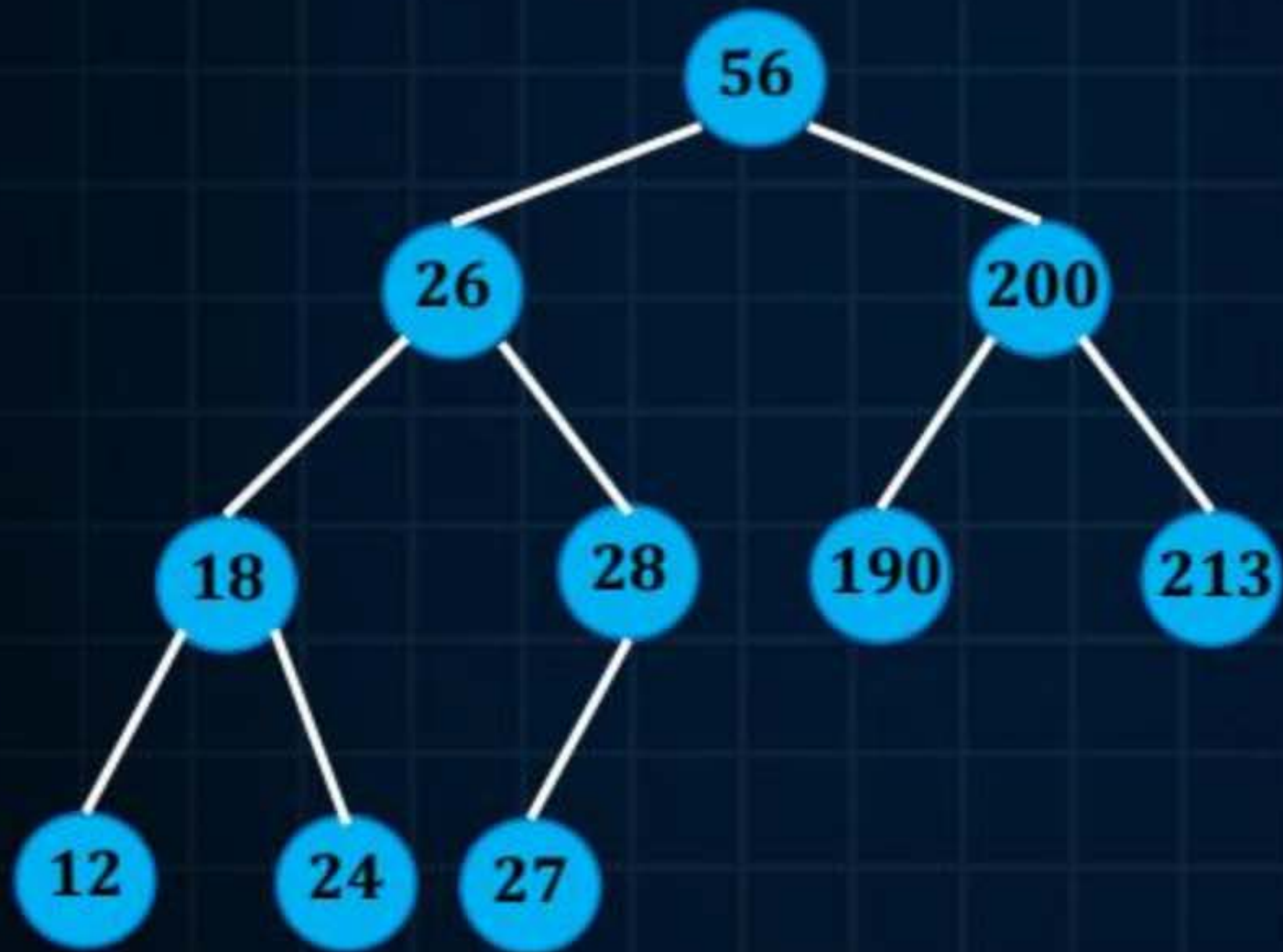
Not sufficient

for construction
of tree uniquely

if 2-ary tree then preorder
and post can construct the uniquely



Binary Search tree



distinct : either left or right subtree

Question

Q. The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)?

Skill

- (A) 2
- (B) 3 ✓
- (C) 4
- (D) 6



if we insert the keys 1, 2, 3, 4, 5, 6, 7, 8 in the

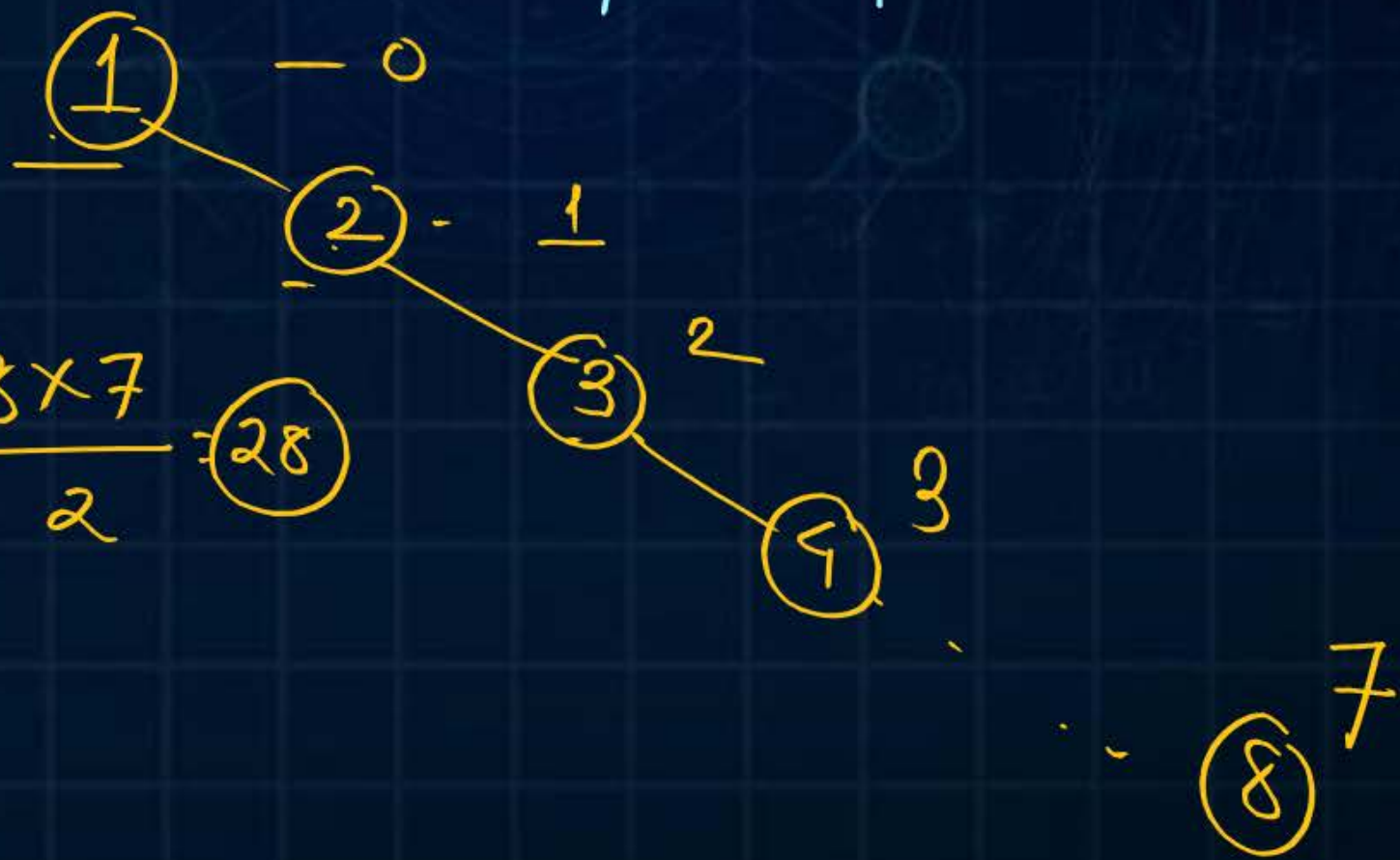
order in Empty BST then No. of comparison Required

for construction of it.

$$\frac{n(n-1)}{2}$$

$$= \frac{8 \times 7}{2}$$

$$= 28$$



1, 2, 3, ..., n — No. of comparison $\frac{n(n-1)}{2}$

$O(n^2)$ worst case

construction cost BST

```
int search ( struct tnode *t, int k) {
```

```
    if (t == NULL)  
        return -1;
```

```
    else if (t->data == k)  
        return 1;
```

```
    else if (t->data > k)
```

```
        return search(t->left, k);
```

```
    else  
        return search(t->right, k);
```

```
}
```

Binary Search tree

Time complexity of

Search in BST

$O(h)$

Balanced Binay Search $O(\log_2 n)$ (AVL tree)

Binay tree $\longrightarrow O(h) \longrightarrow O(n)$ Skewed tree

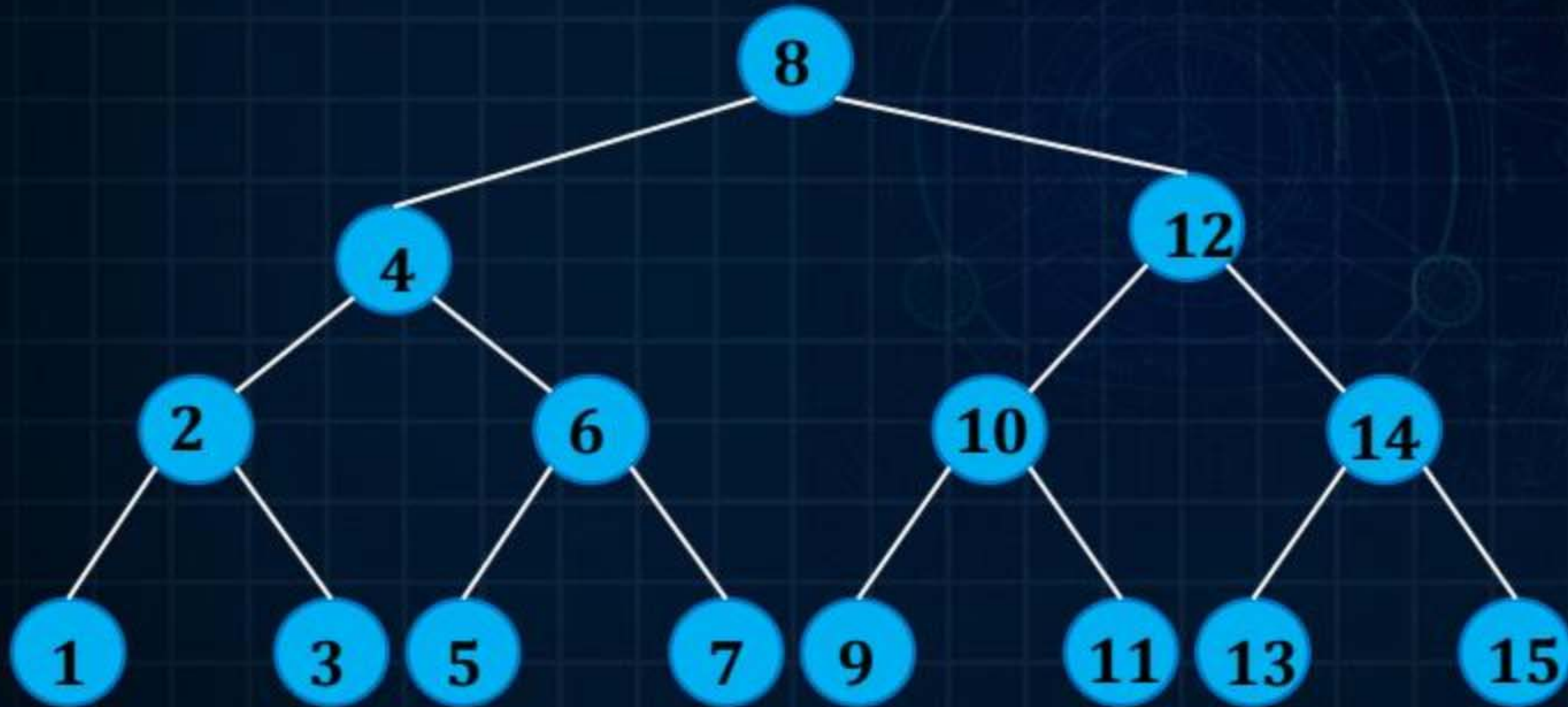
Analysis of Search

- Worst cases *Maximum time Required for the Search Operation is $O(n)$*



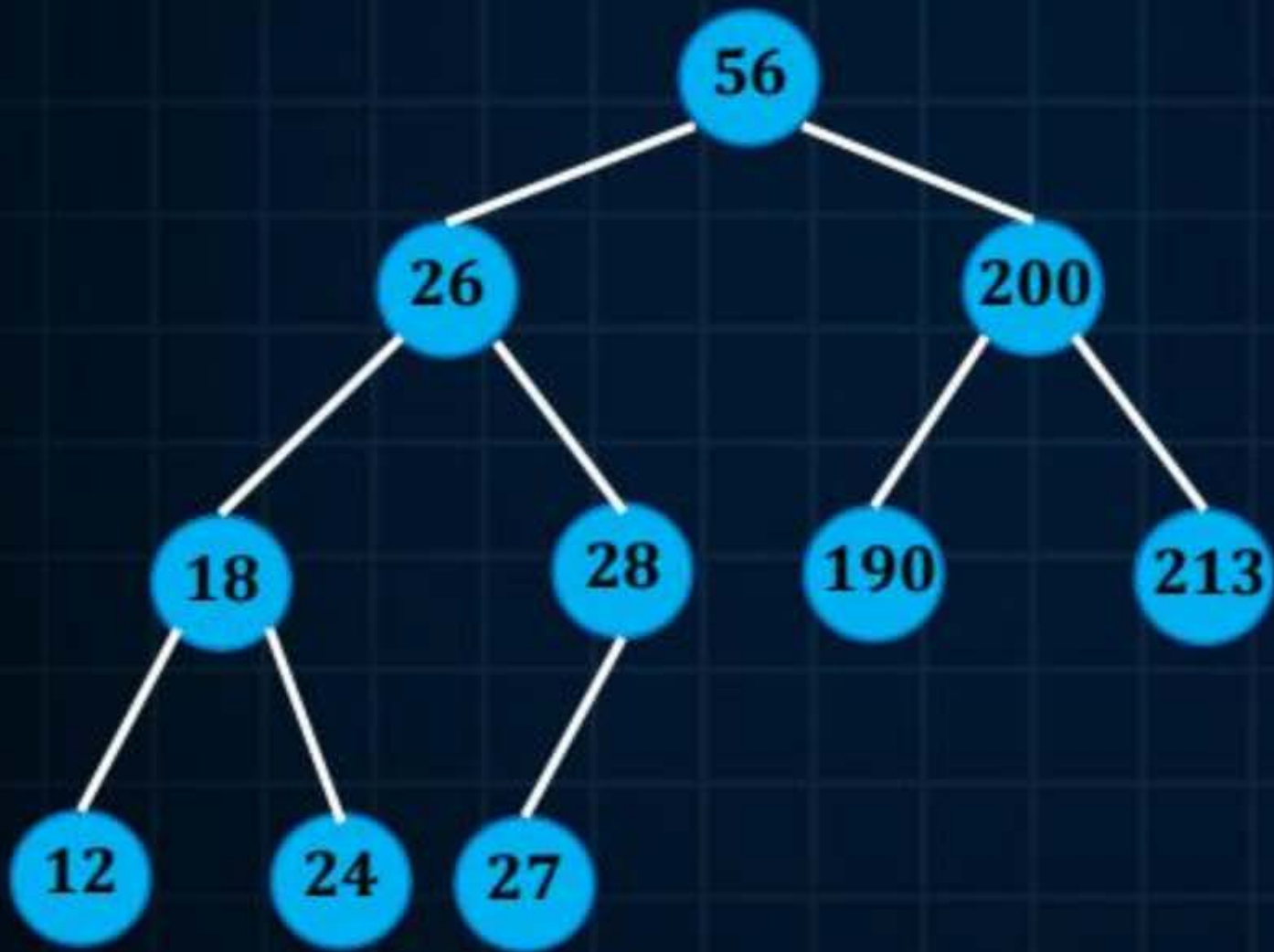
Analysis of Search

- Best cases *Minimum time Required for the Search Operation is $O(\log_2 n)$*





Binary Search tree Traversal



Inorder : 12, 18, 24, 26, 27, 28, 56, 190, 200, 213

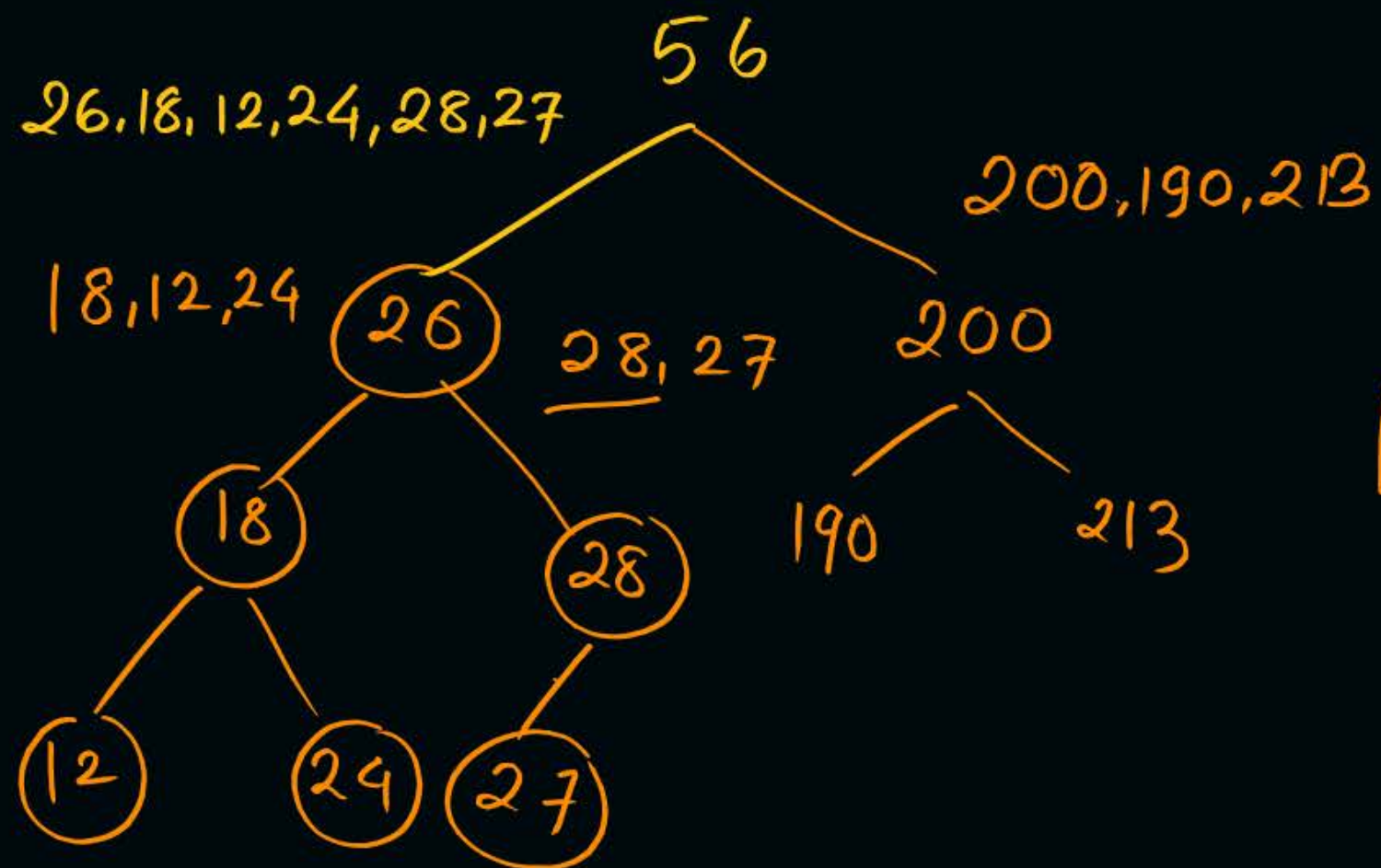
preorder : 56, 26, 18, 12, 24, 28, 27, 200, 190, 213

post-traversal 12, 24, 18, 27, 28, 26, 190, 213, 200
56

Inorder : 12, 18, 24, 26, 27, 28, 56, 190, 200, 213

BST in increasing order Sequence

Root can't be identified



Root left Right

preorder: 56, 26, 18, 12, 24, 28, 27, 200, 190, 213

Only using preorder traversal
tree can be constructed.

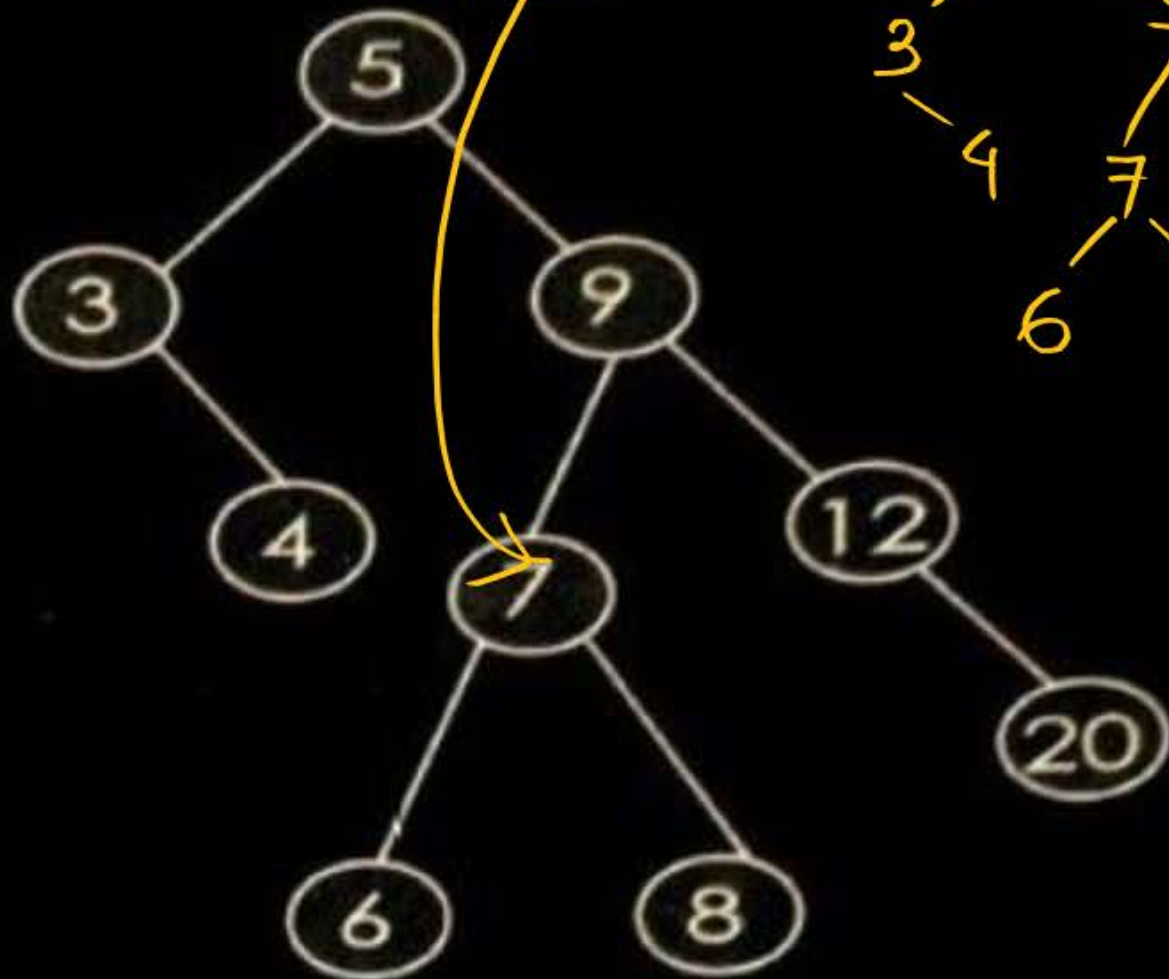


Left Right Root

post-traversal 12, 24, 18, 27, 28, 26, 190, 213, 200
56

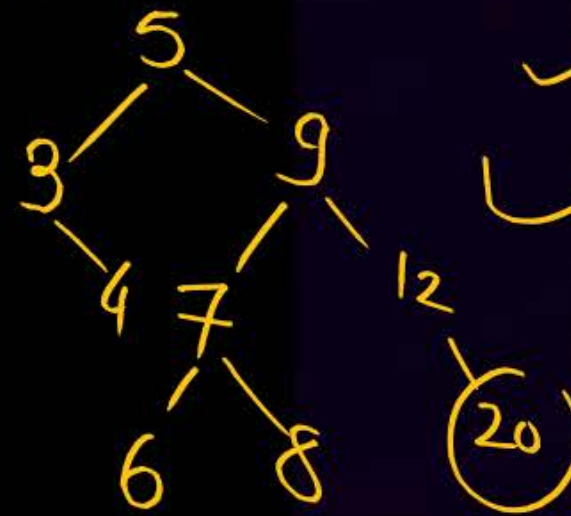
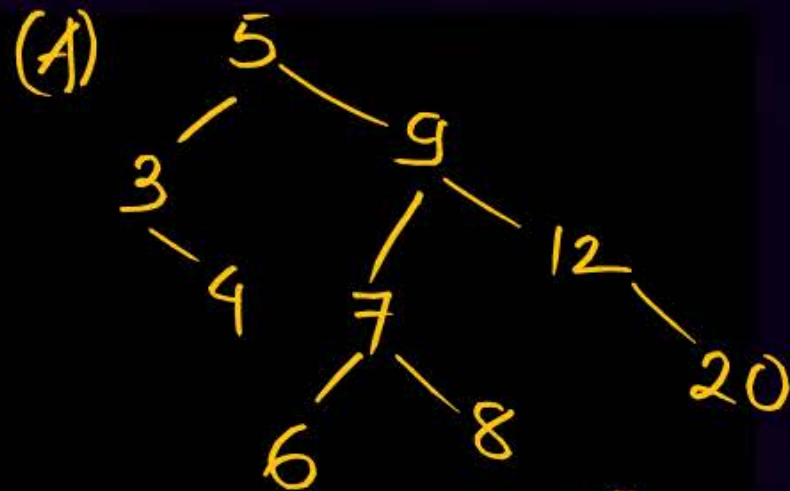
Question

A binary search tree shown below constructed by inserting a sequence of items into empty tree



Not match

MISQ



☒ A

5 3 4 9 12 7 8 6, 20

☒ B

5 9 3 7 6 8 4 12 20

☒ C

5 9 7 3 8 12 6 4 20

☐ D

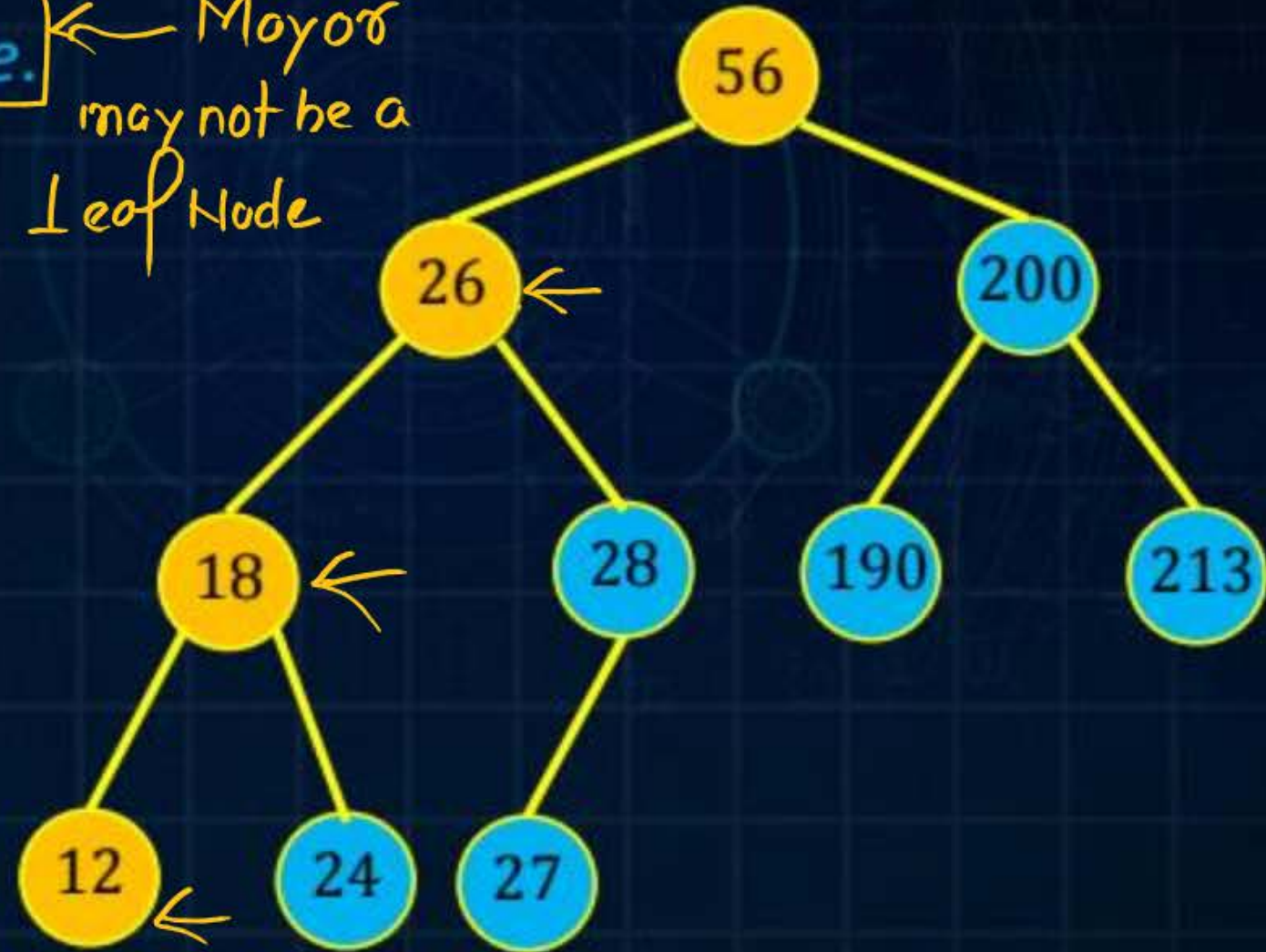
5 9 3 6 7 8 4 12 20

Finding Min & Max

The binary-search-tree property guarantees that: *The minimum is located at the left-most node.*

← Mayor
may not be a
Leaf Node

```
Tree-Minimum(x) {  
  while left[x] ≠ NULL  
    do x = left[x]  
  return x  
}
```



Finding Min & Max

The binary-search-tree property guarantees that: *The maximum is located at the right-most node.*

```
Tree-Maximum(x) {  
    while right[x] ≠ NULL  
        do x = right[x]  
    return x  
}
```

*may or may not be
Leaf Node*



Min Max

Balanced Binay Search $O(\log_2 n)$ (AVL tree)

Binay tree $\longrightarrow O(h) \longrightarrow O(n)$ Skewed tree

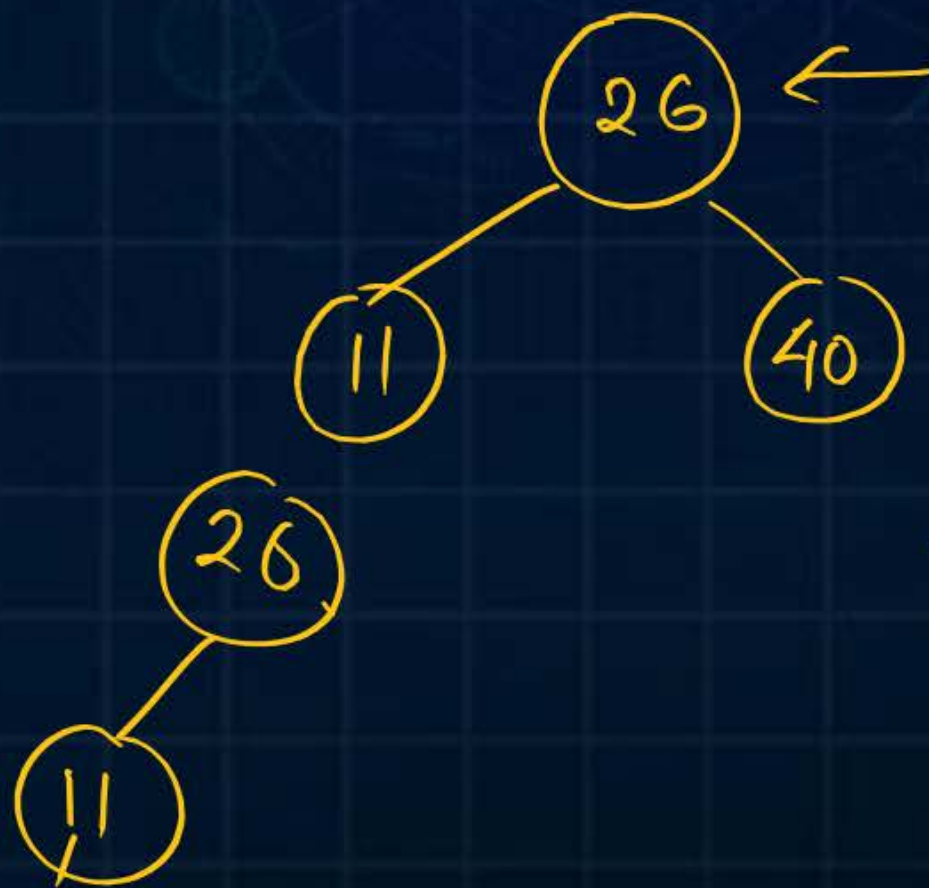
What is the time complexity of Best Algorithm to find an element lesser than max in BST.

(A) $\Theta(\log_2 n)$

(B) $\Theta(n)$

(C) $\Theta(n^2)$

✓ (D) $\Theta(1)$



if right child exists
return $t \rightarrow \text{data}$

No right child
root is max

return $t \rightarrow \text{left}$

if No LC and RC
No value exists



COUNTING BST

No. of BST = Catalan No. $C_n = \frac{1}{n+1} 2^n C_n$

1, 3, 5



Label so that it becomes BST

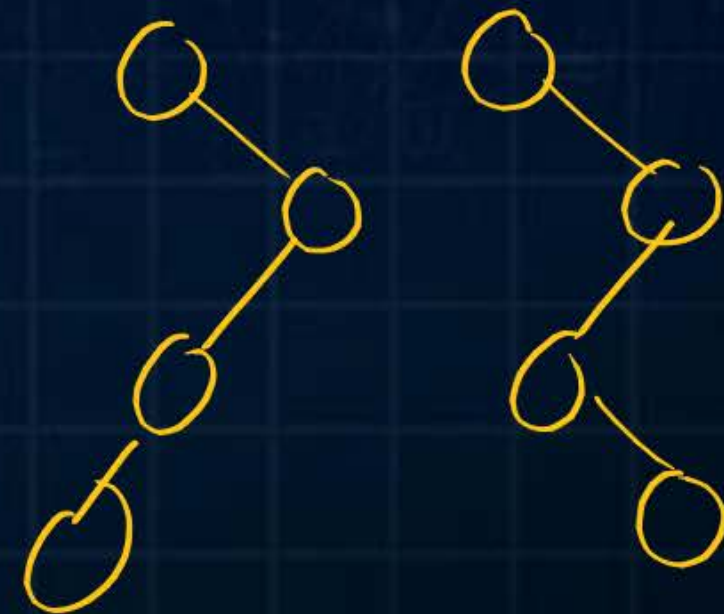
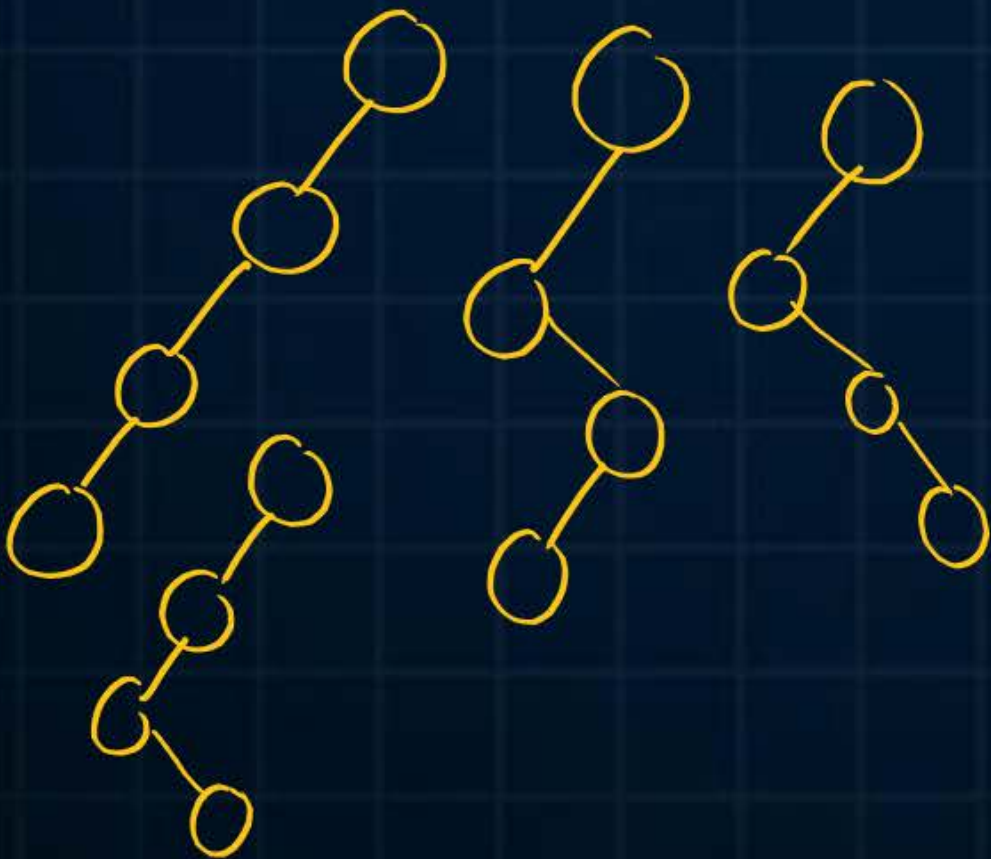
One unlabeled tree can be labeled in only 1 way

Question

Given 4 Distinct element {1,2,3,4}, how many BST will be of height 3.

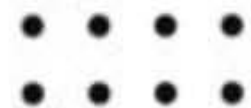
Maximum Height Skewed tree

No. of element distinct
then 2^{n-1} BST
with max height



Binay Heap

Thank
THANK



Keep Hustling!